

Complete the following tables by hand

- For the following tables, fill in all the missing entries by hand.
- Show your work. This may mean adding pages.
- Post the final version back on canvas in one pdf document.

BETA DIVERSITY

Jaccard distance:

a qualitative, non-phylogenetic beta diversity metric

$$J(A, B) = 1 - \frac{A \cap B}{A \cup B}$$

FeatureTable[Frequency]					
	feature1	feature2	feature3	feature4	feature5
gut1	42	0	37	99	1
left.palm1	12	1	22	88	0
right.palm1	25	3	23	86	0
tongue1	0	0	87	12	0



DistanceMatrix				
	gut1	left.palm1	right.palm1	tongue1
gut1	0.0	0.4	0.2	0.6
left.palm1	0.4	0.0	0.2	0.6
right.palm1	0.2	0.2	0.0	0.6
tongue1	0.6	0.6	0.6	0.0

$$4/5 = 0.8, 1 - 0.8 = 0.2$$

$$2/5 = 0.4, 1 - 0.4 = 0.6$$

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$$2/5 = 0.4, 1 - 0.4 = 0.6$$

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BETA DIVERSITY

Bray-Curtis distance:

a quantitative, non-phylogenetic beta diversity metric

FeatureTable[Frequency]					
	feature1	feature2	feature3	feature4	feature5
gut1	42	0	37	99	1
left.palm1	12	1	22	88	0
right.palm1	25	3	23	86	0
tongue1	0	0	87	12	0

$$BC(A, B) = \frac{\sum_i |X_{iA} - X_{iB}|}{\sum_i (X_{iA} + X_{iB})}$$

X_{iA} : frequency of feature i in sample A

DistanceMatrix				
	gut1	left.palm1	right.palm1	tongue1
gut1	0.0	0.19	0.13	0.29
left.palm1	0.19	0.0	-0.05	0.11
right.palm1	0.13	-0.05	0.0	0.16
tongue1	0.29	0.11	0.16	0.0

$$(42-25)+(0-3)+(37-23)+(99-86)+(1-0)/(25+3+23+86+0)+(42+0+37+99+1) = 0.1329$$

$$(42-0)+(0-0)+(37-87)+(99-12)+(1-0)/(0+0+87+12+0)+(42+0+37+99+1) = 0.2877$$

$$(12-25)+(1-3)+(22-23)+(88-86)+(0-0)/(12+1+22+88+0)+(25+3+23+86+0) = -0.054$$

$$(12-0)+(1-0)+(22-87)+(88-12)+(0-0)/(12+1+22+88+0)+(0+0+87+12+0) = 0.108$$

$$(25-0)+(3-0)+(23-87)+(86-12)+(0-0)/(25+3+23+86+0)+(0+0+87+12+0) = 0.161$$

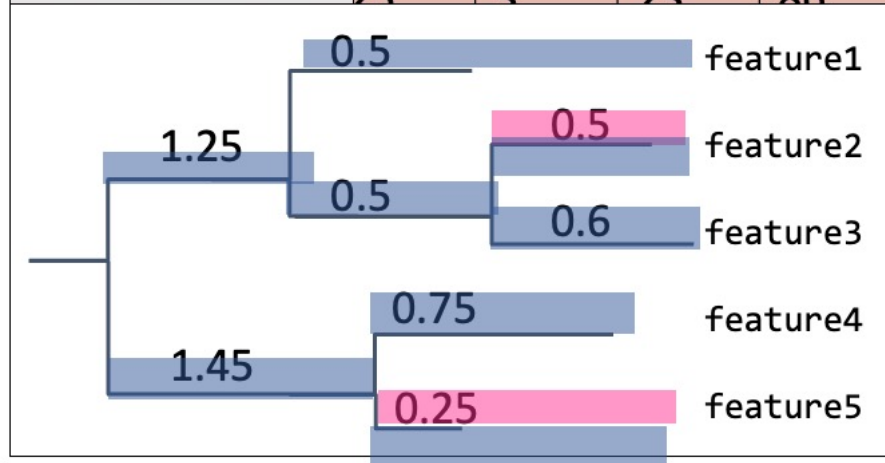
Unweighted UniFrac distance:

$$UU(A, B) = \frac{\text{sum of unique branch length}}{\text{sum of observed branch length}}$$

$$UU(\text{gut1}, \text{leftpalm1}) = (0.5 + 0.25) / (0.5 + 1.25 + 0.5 + 0.5 + 0.6 + 1.45 + 0.75 + 0.25)$$

$$UU(\text{gut1}, \text{leftpalm1}) = 0.75 / 5.8$$

FeatureTable[Frequency]					
	feature1	feature2	feature3	feature4	feature5
gut1	42	0	37	99	1
left.palm1	12	1	22	88	0
Phylogeny[Rooted]	0.5	3	23	86	0



DistanceMatrix				
	gut1	left.palm1	right.palm1	tongue1
gut1	0.0	0.13	0.13	0.23
left.palm1	0.13	0.0	0.04	0.23
right.palm1	0.13	0.04	0.0	0.23
tongue1	0.23	0.23	0.23	0.0

$$1.25+0.5+0.5+0.5+0.6+1.45+0.75+0.25 = 5.8$$

$$0.25 + 0.5 = 0.75$$

$$0.75/5.8 = 0.13$$

$$0.6 + 0.75/5.8 = 0.233$$

$$0.25/5.8 = 0.043$$

$$0.6 + 0.75/5.8 = 0.233$$

$$0.6 + 0.75/5.8 = 0.233$$

Complete clr and rclr by hand for sample 1 & 2 (show work)

ID	feature1	feature2	feature3	feature4	feature5	feature6	feature7
sample1	40	1000	0	30	2	80	18
sample2	10	647	68	150	2025	100	0
sample3	5	208	12	70	200	0	30
sample4	0	20	100	5	5	0	200

Clr transform – replace 0s with small number (use 1 in this example)

ID	feature1	feature2	feature3	feature4	feature5	feature6	feature7
sample1	0.55	3.77	-3.13	0.26	-2.44	1.24	-0.25
sample2	-2.02	2.16	-0.10	0.69	3.30	0.29	-4.32

Rclr transform – Ignore the 0s during the calculation. Fill in after with NA)

ID	feature1	feature2	feature3	feature4	feature5	feature6	feature7
sample1	0.03	3.25	N/A	-0.26	-2.97	0.72	-0.77
sample2	-2.73	1.44	-0.82	-0.03	2.58	-0.43.	N/A

$$7\sqrt[7]{40*1000*1*30*2*80*18} = 23.049$$

$$7\sqrt[7]{10*647*68*150*2025*100*1} = 75.012$$

$$\ln(40/23.049) = 0.55125$$

$$\ln(1000/23.049) = 3.77013289$$

$$\ln(1/23.049) = -3.13762238$$

$$\ln(30/23.049) = 0.263575$$

$$\ln(2/23.049) = -2.4444752$$

$$\ln(80/23.049) = 1.24440425$$

$$\ln(18/23.049) = -0.24725063$$

$$\ln(10/75.012) = -2.01506301$$

$$\ln(647/75.012) = 2.15469819$$

$$\ln(68/75.012) = -0.0981404$$

$$\ln(150/75.012) = 0.69298719$$

$$\ln(2025/75.012) = 3.29567688$$

$$\ln(100/75.012) = 0.28752209$$

$$\ln(1/75.012) = -4.3176481$$

$$6\sqrt[6]{40*1000*1*30*2*80*18} = 38.8832 \sim 38.883$$

$$6\sqrt[6]{10*647*68*150*2025*100*1} = 154.04759 \sim 154.0476$$

$$\ln(40/38.883) = 0.02832232$$

$$\ln(1000/38.883) = 3.24719814$$

$$\ln(30/38.883) = -0.25935976$$

$$\ln(2/38.883) = -2.96740996$$

$$\ln(80/38.883) = 0.7214695$$

$$\ln(18/38.883) = -0.77018538$$

$$\ln(10/154.0476) = -2.73467655$$

$$\ln(647/154.0476) = 1.43508465$$

$$\ln(68/154.0476) = -0.81775394$$

$$\ln(150/154.0476) = -0.02662635$$

$$\ln(2025/154.0476) = 2.57606333$$

$$\ln(100/154.0476) = -0.43209146$$

Calculate Euclidean Distance

- For the clr (or rclr – replace NA with smallest number in dataset) calculate the Euclidean Distance between samples 1 and 2

In general, for points given by Cartesian coordinates in n -dimensional Euclidean space, the distance is^[6]

$$d(p, q) = \sqrt{(p_1 - q_1)^2 + (p_2 - q_2)^2 + \cdots + (p_n - q_n)^2}.$$

0.55	3.77	-3.13	0.26	-2.44	1.24	-0.25
-2.02	2.16	-0.10	0.69	3.30	0.29	-4.32

$\sqrt{(0.55 - (-2.02))^2 + (3.77 - 2.16)^2 + (-3.13 - (-0.10))^2 + (0.26 - 0.69)^2 + (-2.44 - 3.30)^2 + (1.24 - 0.29)^2 + (-0.25 - (-4.32))^2}$
 $\sqrt{58.8425} = 7.67088652 \sim$ The euclidean distance between samples 1 & 2 is approximately 7.67